Antoine Vuignier

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I am completing my PhD in theoretical physics at EPFL, with a defense scheduled in August 2026. My work focuses on turning abstract questions into well-defined mathematical problems and producing quantitative answers, using a combination of analytical reasoning and computational methods. Alongside research, I have accumulated significant experience teaching and supervising students at university level. I am looking to join an applied research environment in Valais where I can contribute both to project-driven scientific work and to student supervision and teaching.

Education

2022 – ... ◇ **PhD, EPFL - Fields and Strings Laboratory**
Research on non-perturbative quantum field theory and quantum gravity: S-matrix bootstrap (bounding scattering amplitudes via semidefinite programming) and the large-N matrix model approach to emergent spacetime. Computational workflows in Python and Wolfram Language on HPC clusters (SDPB, SLURM, spectral methods).

2017 – 2022 ◇ **Bachelor and Master of Physics, EPFL**

Teaching and scientific activities

- ◇ **Teaching Assistant, EPFL** - 10+ courses on physics and mathematics at the bachelor and master level. Supervised two master students on their thesis. Gave lectures, designed projects and supervised students for EPFL summer schools.
- ◇ **Peer Reviewer, JHEP** - Served as peer reviewer for JHEP, evaluating manuscripts on quantum gravity for scientific rigor, methodology, and novelty.
- ◇ **Research seminar organizer, EPFL** End-to-end coordination of weekly seminar series, including speaker selection, logistics, chairing, and communication.

Skills

Coding ◇ Python (NumPy, SciPy, PyTorch, JAX/Flax/Optax, matplotlib), Wolfram Language, LaTeX, C++. HPC cluster workflows (SLURM, Docker), semidefinite programming (SDPB).

Languages ◇ Native in French, fluent in English, B2 in German.

General ◇ Scientific communication, teaching and student supervision, project coordination.

Research Output

- ◇ 5 publications in high energy theoretical physics (JHEP), see INSPIRE profile `https://inspirehep.net/authors/2612961`.
- ◇ Multiple invited talks at international workshops and universities

Scientific Computing & Numerical Methods

- ◇ **Large-scale optimization pipelines**- Developed and ran semidefinite programming (SDP) workflows on HPC clusters (SDPB, SLURM, custom Python preprocessing) to derive rigorous numerical bounds in quantum field theory. This is the core computational method of my PhD — constrained optimization at scale over high-dimensional function spaces.
- ◇ **Pseudospectral PDE solver**- Implemented a Chebyshev collocation solver for a 2D Schrödinger equation with a strongly anisotropic potential. Used domain compactification, coordinate rescaling, symmetry reduction, and Kronecker-product Hamiltonian assembly. Verified convergence across grid sizes M=22–48. Code supports published results in JHEP 2025.
- ◇ **Boundary value problem solver**- Solved a singular 4D electrostatic integral equation via Legendre basis expansion and adaptive quadrature, reducing to a linear least-squares system. Supports numerical results in arXiv:2411.18678.
- ◇ **Variational numerical solver**- Reformulated a physics optimization problem (minimizing an effective action over eigenvalue densities in a large-N matrix model) as a neural network training loop, using an MLP as a function approximator. Recovered analytic benchmarks and captured phase transitions. Implemented in PyTorch and JAX.